Classwork 04//LSTM

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Due Date:** Feb 19, 2026

Assume

- W_f, b_f weights and bias for the forget gate, U_f weights for recurrent connection for the forget gate.
- W_i, b_i weights and bias for the input gate, U_i weights for recurrent connection for the input gate.
- W_o, b_o weights and bias for the output gate, U_o weights for recurrent connection for the output gate.
- c_t is the cell state or the context vector at time t (like a memory tape).
- W_c, b_c weights and bias for the cell state, U_c weights for recurrent connection for the cell state.

Forget gate operation:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

Input gate operation:

$$i_i = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

Updated cell state:

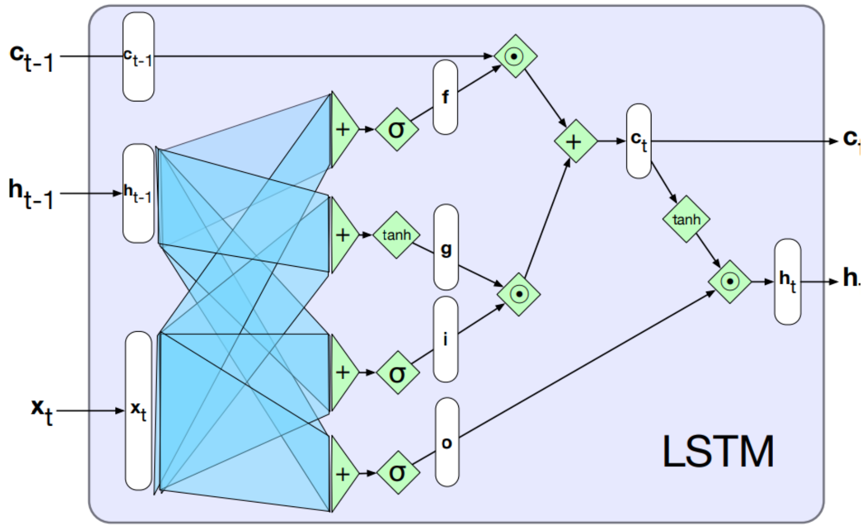
$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$
$$c_t = f_t \odot c_{t-1} + i_t \cdot \tilde{c}_t$$

Output gate:

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

Hidden state:

$$h_t = o_t \odot \tanh(c_t)$$



1 LSTM Output Calculation

Initial input $x_t = [0.5]$
Previous hidden state $h_{t-1} = [0.2]$
Previous cell state $c_{t-1} = [0.8]$

1. For the forget gate, consider $W_f = [0.5]$, $U_f = [0.3]$, $b_f = 0.0$ Compute the forget gate output f_t.

2. For the input gate, consider $W_i = [0.6]$, $U_i = [0.4]$, $b_i = 0.0$ Compute the input gate output i_t.

3. For the candidate cell, consider $W_c = [0.4]$, $U_c = [0.5]$, $b_c = 0.0$ Compute the updated cell state c_t.

   What percentage of old memory is kept?

   What percentage of new candidate is kept?

4. For the output gate, consider $W_o = [0.7]$, $U_o = [0.6]$, $b_o = 0.0$ Compute the input gate output o_t.

5. Compute the final hidden state h_t.

1.1 Solution

1. **Forget gate computation:**

   $W_f x_t + U_f h_{t-1} + b_f = 0.5 \times 0.5 + 0.3 \times 0.2 + 0.0 = 0.31$

   $f_t = \frac{1}{1+exp(-0.31)} = 0.5769$

2. $W_i x_t + U_i h_{t-1} + b_i = 0.6 \times 0.5 + 0.4 \times 0.2 + 0.0 = 0.38$
   $i_t = \frac{1}{1+exp(-0.38)} = 0.5939$

3. $W_c x_t + U_c h_{t-1} + b_c = 0.4 \times 0.5 + 0.5 \times 0.2 + 0.0 = 0.30$
   $\tilde{c}_t = \tanh(0.30) = 0.2913$
   $c_t = 0.5769 \times 0.8 + 0.5939 \times 0.2913 = 0.4615 + 0.1730 = 0.6345$
   Based on $f_t = 0.5769$, roughly 57.69% of the older memory is kept.
   Based on $i_t = 0.5939$, roughly 59.39% of the new candidate is kept.

4. $W_o x_t + U_o h_{t-1} + b_o = 0.7 \times 0.5 + 0.6 \times 0.2 + 0.0 = 0.35 + 0.18 = 0.53$
   $o_t = \frac{1}{1+exp(-0.53)} = 0.6295$

5. $h_t = 0.6295 \times \tanh(0.6345) = 0.3532$.